MILITARY SPECIFICATION SHEET

ELECTRON TUBE, GAS SWITCHING

TYPE 6560

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Dual TR tube, broadband, fixed tuned, frequency range of 15,000 to 17,000 MHz

ABSOLUTE RATINGS:

Parameter:	Incident power	Ebb	prı	Alt	Du
Unit:	kw	Vdc	pps	ft	---
Maximum:	100	---	3,000	10,000	0.0015
Minimum:	1	-700	---	---	---

PHYSICAL CHARACTERISTICS:

Dimensions: See figure 1
Weight: 3 ounces

TEST CONDITIONS:

Parameter:	Incident power	tp1	tp2	prr	Ii	F
Unit:	kw	μs	μs	pps	μAdc	MHz
Tolerance:	±10%	±0.05	±0.025	---	---	---
Test condition 1:	20	0.5	0.25	3,000	100	F3
Test condition 2:	20	0.5	---	3,000	100	F3
Test condition 3:	90	0.5	---	3,000	100	F4

FREQUENCY		
F	MHz	±
1	15,000	0.1%
2	15,500	0.1%
3	16,000	0.1%
4	16,000	5 %
5	16,500	0.1%
6	17,000	0.1%

GENERAL:

Qualification - Required

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METHOD	REQUIREMENT OR TEST	NOTES	TEST	CONDITIONS	SYMBOL	LIMITS		UNIT
						MINIMUM	MAXIMUM	
	Quality conformance inspection, part 1	1	-					
1027	Temperature cycling (nonoperating)	-	-	1 cycle	---	---	---	---
4406	Ignitor voltage drop	2	-	Ii = 100 μAdc	Eid	200	475	Vdc
4452	Spike-leakage energy	-	1		Ws	---	0.10	erg
4452	Flat-leakage power	-	1		pf	---	20	mw
4489	Duplexer loss	3	-	Ii = 100 μAdc				
				F = F1	---	---	1.4	dB
				F = F3	---	---	1.2	dB
				F = F5	---	---	1.4	dB
4473	Low-level VSWR	3	-	F = F1	σ	---	1.4	---
				F = F2	σ	---	1.2	---
				F = F3	σ	---	1.2	---
				F = F4	σ	---	1.2	---
				F = F5	σ	---	1.4	---
	Quality conformance inspection, part 2							
4003	Pressurizing	-	-		---	---	30	lbs (gage)
4471	Recovery time	-	3		t	---	10	μs
4489	Duplexer loss	-	-	F = F1 thru F5; Ii = 100 μAdc	---	---	1.2	dB
4401	Ignitor ignition time	2	-	Ebb = -700 Vdc; R = 2.7 Meg	t	---	5	sec
4488	Arc loss	-	2		La	---	0.8	dB
4101	Dielectric material strain	-	-		---	---	---	---
	Quality conformance inspection, part 3							
---	Life test	-	3	Group D	t	500	---	hrs
---	Life test end points:							
4452	Spike-leakage energy	-	1		Ws	---	0.15	erg
4452	Flat-leakage power	-	1		pf	---	20	mw
4471	Recovery time	-	3		t	---	20	μs
4489	Duplexer loss	-	-	F = F3; Ii = 100 μAdc	---	---	1.2	dB

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METHOD	REQUIREMENT OR TEST	NOTES	TEST	CONDITIONS	SYMBOL	LIMITS		UNIT
						MINIMUM	MAXIMUM	
	Periodic-check tests							
4491	Transmitter-receiver isolation	-	-	Ii = 100 μAdc				
				F = F1	---	10	---	dB
				F = F2	---	12	---	dB
				F = F3	---	12	---	dB
				F = F4	---	12	---	dB
				F = F5	---	10	---	dB
1027	Temperature cycling life test	-	-	10 cycles	---	---	---	---
4021	Degradation due to vibration	-	-	F = 50 ±2Hz; 10 G	---	---	---	---
4474	High-level VSWR	-	2		σ	---	1.2	---

NOTES:

1. Unless otherwise specified, the AQL for all tests listed under quality conformance inspection, part 1, shall be 1.0, inspection level II.
2. Ignitor ignition time and ignitor voltage drop shall be measured separately for each ignitor electrode.
3. A swept frequency method may be used for this test.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Air Force - 11, 17
DSA - ES

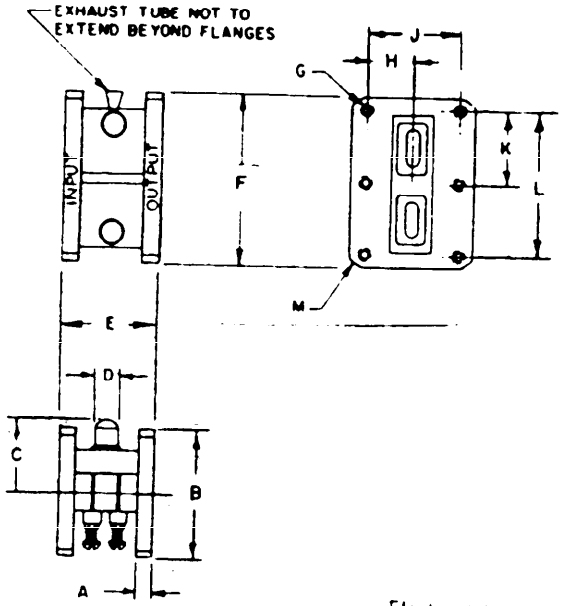
User activities:
Army - AV, WC
Navy - AS, OS, CG, SH

Preparing activity: Navy - EC

Agent: DSA - ES

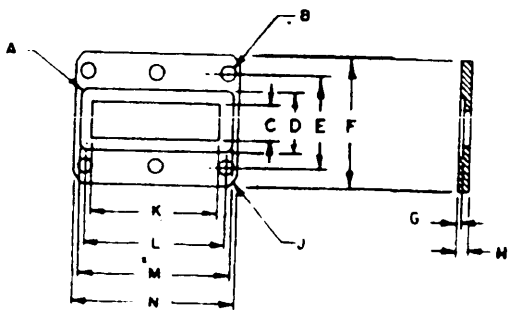
(Project 5960-2632)

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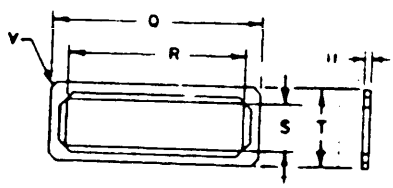
Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 1		
E	.797 (20.24)	.817 (20.75)
G	No. 6-32UNC-2B 6 holes ea flange	
Quality conformance inspection, part 2		
C		.875 (22.23)
H	.476 (12.09)	.480 (12.19)
J	.951 (24.16)	.961 (24.41)
K	.745 (18.92)	.755 (19.18)
L	1.490 (37.85)	1.510 (38.35)
Quality conformance inspection, part 3 (periodic check)		
A	.083 (2.11)	.103 (2.62)
B	1.297 (32.94)	1.327 (33.71)
D	.245(6.22)	.255(6.48)
F	1.735 (44.07)	1.765 (44.83)
M	.875 (22.23)	

Electron tube type 6560.



Ref	Dimensions
A	.06 (1.52) ±.016 (.41)
B	No. 26, .147 (3.73) dia, 6 holes
C	.394 +.002, −.000 (10.01 +.05, −.00)
D	.656 (16.66)
E	.956 ±.004 (24.28 ±.10)
F	1.31 (33.27) ±.016 (.41)
G	.023 ±.001 (.58 ±.03)
H	.09 (2.29) ±.016 (.41)
J	.88 (22.35) ±.016 (.41)
K	1.364 +.002, −.000 (34.65 +.05, −.00)
L	1.500 ±.004 (38.10 ±.10)
M	1.625 (41.28)
N	1.75 (44.45) ±.016 (.41)

Mating flange.



Ref	Dimensions
Q	1.58 (40.13) ±.016 (.41)
R	1.384 ±.005 (35.15 ±.13)
S	.431±.005 (10.95 ±.13)
T	.61 (15.49) ±.016 (.41)
U	.037±.005 (.94 ±.13)
V	.03 (.76) X 45° ±.016(.41)

Gasket.

NOTE
Rhodium flash over silver plating is optional.

FIGURE 1. Outline drawing of electron tube type 6560.

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NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-28254

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